PREP 2 · LESSONS 3–4

Drop tests and ramps

Repeats and means · bar charts · choosing a scale · plotting a line graph

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define accuracy. DN 851 [1]

2 What is the independent variable? DN 870 [1]

3 Name the type of graph used when the independent variable is continuous. DN 882 [1]

Key ideas LESSONS 3–4

Show your working

4 A ball is dropped onto three materials. The table shows its bounce height.

Material	Trial 1 / cm	Trial 2 / cm	Trial 3 / cm	Mean / cm
Foam	12	14	13	13
Bubble wrap	20	23	20	
Carpet	31	28	31	30

a Calculate the mean bounce height for bubble wrap. Write it in the table. [1]

b Which material is the best shock absorber? Give the reason. [1]

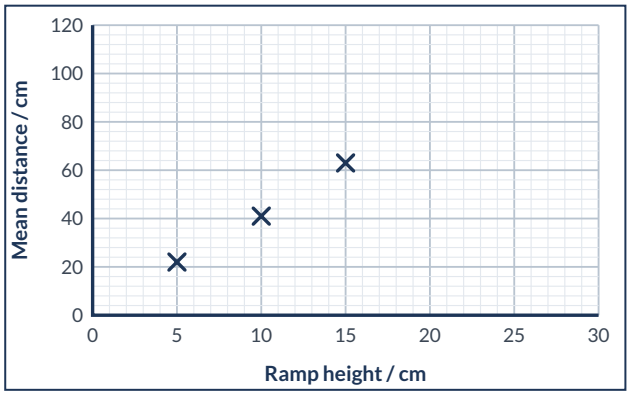
c Explain why these results are shown on a bar chart. [1]

5 In the ramp practical, a trolley leaves a ramp and travels across the floor.

Ramp height / cm	5	10	15	20	25
Mean distance / cm	22	41	63	80	101

a Plot the last two points on the graph. [2]

b On a different graph, the largest value on the y-axis is 118 cm and there are 10 big squares. Draw a ring around the best value for each big square: [1]

10 cm	15 cm	20 cm	50 cm

c Name one control variable in the ramp practical. [1]